Sahaj Bhadja

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Summary

Systems-focused software engineer with 2+ years building C++ and Python software, simulation pipelines, concurrent workflows, and cross-platform services. Strong foundation in operating systems, algorithms, deterministic execution, verification, debugging, performance analysis, and reliable software delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built performance-sensitive Python services for 500+ users, profiling execution paths and optimizing indexed queries, schemas, and caching to reduce response latency from 1.2 seconds to 350 milliseconds.
- Engineered fault-tolerant concurrent workflows with explicit states, bounded execution, failure isolation, automated tests, and recovery, tripling throughput while improving deterministic runtime behavior, operational observability, and production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Python services handling 10,000+ daily requests, tracing failures through logs, source code, state transitions, and error paths while resolving reliability issues affecting more than 150 users.
- Engineered multithreaded processing for 1,000+ files with bounded workers, automated validation, progress instrumentation, and failure recovery, reducing turnaround 60% while maintaining stable behavior under concurrent workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built autonomous-driving simulation and model-evaluation pipelines over 100,000+ trajectory records, improving planning accuracy 18% through reproducible experiments, controlled scenarios, performance monitoring, and systematic software defect analysis.
- Developed a spatial-temporal encoder integrating map geometry and agent context, improving closed-loop planning stability 15% while validating outputs through repeatable simulation rollouts and clearly documented quantitative analysis.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed modular Python services and data-processing integrations using AWS Glue, S3, SQL, pagination, and caching, improving response times 30% and processing efficiency 68% across demanding production applications.
- Maintained 20+ unit and integration suites with Jenkins, PyTest, and Mockito, accelerating software defect isolation, regression verification, and dependable releases across Windows, Linux, and macOS production environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular services across eight microservices with authentication, WebSocket messaging, state synchronization, and defensive error handling, securely processing more than two million records through documented software interfaces.
- Automated functional tests, staged deployments, and production diagnostics through Jenkins, Docker, and Azure, reducing release time 50% while improving system validation coverage, deployment consistency, and operational visibility.

Skills

Languages & Fundamentals: C++, Python, Java, SQL, object-oriented programming, data structures, algorithms, operating systems, computer architecture fundamentals, deterministic systems concepts
Simulation & Autonomy: Autonomous-driving simulation, trajectory analysis, planning evaluation, PyTorch, closed-loop testing, controlled scenarios, model validation, robotics concepts, state processing
Systems & Quality: Multithreading, concurrency, Linux, Windows, macOS, source-level debugging, performance profiling, unit testing, integration testing, system testing, PyTest, Mockito, failure isolation
Delivery & Tooling: Git, Jenkins, CI/CD, Docker, Azure, AWS, code reviews, requirements analysis, technical documentation, monitoring, regression testing, production support, Visual Studio familiarity

Projects

CPU Scheduling Simulator | Live | *C++, React, Algorithms, Interactive Visualization*
- Designed and built an interactive C++ scheduling simulator with React controls and Gantt visualization, validating edge cases and explaining performance tradeoffs across more than 1,000 process combinations.

Chat2SQL | Code | *Java, TypeScript, SQL, Docker, Maven*
- Built a modular Java and TypeScript tool supporting 10+ database engines with schema-aware execution, input validation, error handling, automated tests, and Docker-backed reproducible deployment across multiple environments.

FinNexus | Live | *Real-Time Systems, WebSockets, Kubernetes, Monitoring*
- Built a real-time platform supporting 50,000+ users and one million transactions, integrating WebSocket updates, failure monitoring, Kubernetes deployment, automated releases, and performance-tuned services under high-volume workloads.